Program 8

Aim:

Install langchain, cohere (for key), langchain-community. Get the api key(By logging into Cohere and obtaining the cohere key). Load a text document from your google drive . Create a prompt template to display the output in a particular manner.

Theory:

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LangChain is an open-source framework used to build applications powered by large language models (LLMs). It helps developers integrate language models with external data sources such as documents, databases, or APIs. LangChain provides tools like document loaders, prompt templates, and chains to process data and generate meaningful outputs.

In this experiment, the required libraries such as langchain, cohere, and langchain-community are first installed. The Cohere API key is obtained by logging in to the Cohere platform and generating one from the dashboard. This API key allows the program to access Cohere's language models.

Next, a text document stored in Google Drive is loaded using a document loader from the LangChain community package. The document content is then processed and passed to a language model.

A Prompt Template is created to structure how the input is given to the model and how the output should be generated. The prompt template defines a specific format (for example: summary, key points, or conclusion) so that the model produces results in a consistent and readable manner.

Thus, by combining LangChain with Cohere and using a prompt template, we can load documents, process their content, and generate structured outputs using large language models.

Steps of the Program

1. Install Required Libraries
   Install the necessary libraries such as langchain, cohere, and langchain-community using pip.

2. Import Required Packages
   Import the required Python modules like os, TextLoader, PromptTemplate, and the Cohere model from LangChain.

3. Get the Cohere API Key
   Log in to the Cohere dashboard, generate an API key, and store it in the program using an environment variable.

4. Mount Google Drive
   Connect Google Drive to Google Colab so that files stored in the drive can be accessed.

5. Load the Text Document
   Use TextLoader from langchain-community to load the text document (e.g., sample.txt) from Google Drive.
6. Extract the Document Content
   Read the loaded document and store its content in a variable to be processed by the language model.
7. Initialize the Cohere Language Model
   Create an instance of the Cohere model in LangChain using the API key.
8. Create a Prompt Template
   Define a PromptTemplate to specify how the input text should be processed and how the output should be formatted (e.g., summary, key points, conclusion).
9. Create a Processing Chain
   Combine the prompt template and the language model to create a pipeline that processes the input text.
10. Generate the Output
    Pass the document text to the chain and print the generated output in the specified format.

```
!pip install -q langchain langchain-community langchain-core langchain-cohere cohere
```

```
from google.colab import files

uploaded = files.upload()
filename = list(uploaded.keys())[0]

print("Uploaded file:", filename)
```

Choose files sample.txt

sample.txt(text/plain) - 1480 bytes, last modified: 05/04/2026 - 100% done

```
Saving sample.txt to sample (4).txt
Uploaded file: sample (4).txt
```

```python
from langchain_community.document_loaders import TextLoader

loader = TextLoader(filename)
documents = loader.load()

text_data = documents[0].page_content

print(text_data)
```

```
Sample Text (sample.txt)

Artificial Intelligence (AI) is a branch of computer science that focuses on creating mach

Machine Learning is a subfield of AI that enables computers to learn from data without bei

Deep Learning is a specialized area of machine learning that uses neural networks with man

Artificial Intelligence is transforming many industries, including healthcare, agriculture

In the future, AI will continue to evolve and play an important role in solving complex gl
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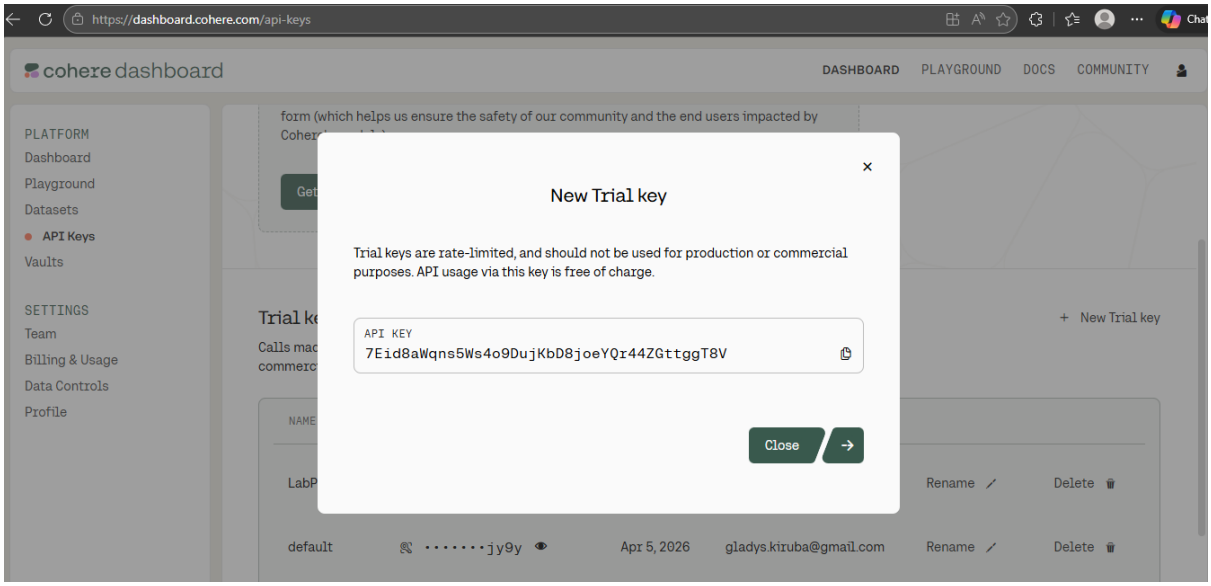
Steps to Get the Cohere API Key

1. Open the Cohere Website
   Go to the official website: https://cohere.com
2. Sign Up or Log In
   - If you are a new user, click Sign Up and create an account using email or Google.
   - If you already have an account, click Log In.
3. Open the Cohere Dashboard
   After logging in, you will be redirected to the Cohere Dashboard.
4. Go to the API Keys Section
   In the dashboard menu, click API Keys.
5. Create a New API Key
   Click the Create API Key button. A new key will be generated.
6. Copy the API Key
   Copy the generated API key. This key is required to access Cohere language models in your program.
7. Use the API Key in Your Code
   Store the key in your Python code using an environment variable:




```python
import os
from langchain_cohere import ChatCohere

os.environ["COHERE_API_KEY"] = "7Eid8aWqns5Ws4o9DujKbD8joeYQr44ZGttggT8V"

llm = ChatCohere(
    cohere_api_key=os.environ["COHERE_API_KEY"],
    model="command-r-08-2024",
    temperature=0.5
)
```

```python
prompt = ChatPromptTemplate.from_template("""
Read the following document and generate output in this format:

Summary:
Key Points:
Conclusion:

Document:
{text}
""")

chain = prompt | llm

result = chain.invoke({"text": text_data})

print(result.content)
```

```
Summary:
Artificial Intelligence (AI) and its subfields, Machine Learning and Deep Learning, are revolutioniz

Key Points:
- AI aims to create intelligent machines capable of performing complex tasks like learning, reasonin
- Machine Learning allows computers to learn from data and make predictions without explicit program
- Deep Learning, a specialized form of Machine Learning, uses neural networks to model complex data
- AI is transforming industries such as healthcare, agriculture, transportation, finance, and educat
- The future of AI is promising, but ethical considerations regarding data privacy, fairness, and tr

Conclusion:
The potential of AI to solve complex problems and improve various aspects of our lives is immense. H
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